

Assignment 2 – Individual

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1. Executive summary

The consumer regulator manages a large volume of complaint records across companies, sectors, and regions, creating increasing pressure to identify risk earlier and allocate compliance resources more effectively. While complaint data captures important operational signals, it does not currently provide a clear business view of which company characteristics are linked to stronger fitness performance or whether certain complaint issues repeatedly emerge together. Without this visibility, regulatory actions may remain reactive rather than preventative.

This report was developed to support stronger regulatory oversight by identifying the business conditions associated with company fitness outcomes, building a reliable fitness assessment capability, and detecting recurring complaint combinations that may indicate broader operational problems.

The analysis found that customer satisfaction levels and complaint recency are the strongest business indicators of company fitness (see Section 2.2). Companies with stronger customer ratings and longer complaint-free periods generally achieved better fitness outcomes, indicating that both service quality and complaint history are important signals of business stability. The analysis also identified clear complaint concentration in metropolitan commercial areas and service-heavy industries such as finance and hospitality (see Section 2.3), showing where regulatory attention is likely to have the greatest impact.

To strengthen future assessment capability, two predictive models were developed to estimate company fitness scores using operational and complaint-related information (see Section 3). The comparative evaluation showed that the Random Forest model delivered the strongest overall predictive accuracy and consistency when compared with the Linear Regression model (see Table 1, Section 4.1). Based on this result, the Random Forest model is recommended as the primary operational tool for future company fitness assessment, while the Linear Regression model remains useful for explaining key business drivers behind fitness outcomes.

The complaint co-occurrence analysis identified three recurring complaint ecosystems across companies, with the strongest involving complaint codes CC-721, CC-774, and CC-921, followed by two additional recurring patterns centred around CC-834 and CC-361 (see Section 4.3). These findings show that complaint issues often emerge in connected clusters rather than as isolated incidents, indicating broader operational weaknesses. This provides the regulator with a stronger basis for targeted policy intervention by focusing on systemic complaint patterns rather than individual complaint cases.

Based on the findings, three actions are recommended. First, integrate predictive fitness scoring into complaint review processes to improve risk prioritisation. Second, strengthen compliance monitoring in sectors and regions with higher complaint concentration. Third, develop cluster-based policy responses targeting the identified complaint groups to address underlying issue before they escalate. Together, these actions can improve regulatory efficiency, support earlier identification of business risk, and strengthen preventative intervention strategies to reduce repeated complaint escalation.

2. Data exploration, pattern discovery, and preparation

2.1 Data Preparation and Transformation

The analytical process began by joining the company and complaint datasets using Company_ID as the common key, creating a unified complaint-level dataset for exploration and predictive modelling. Following joining, predictor attributes were selected based on their likely influence on regulatory fitness performance. These included Region, Sector, Complaint_Code, Avg_Cust_Rating_ATC, Avg_Cust_Trans_Day_ATC, Months_Last_Complaint_ATC, and Num_Staff_ATC. The target variable for predictive modelling was defined as Score_ATC, the regulator's assessed fitness score, as it aligns with the business objective. Company_ID was set as ID role. As Complaint_Description is omitted, it was deselected from the process.

Data quality preparation included duplicate detection and removal to prevent repeated complaint records, conducted through Remove Duplicates operator. Invalid and zero-value entries in selected fields (Region, Sector, Num_Staff_ATC) were also treated as missing values. Missing values were handled by removing records to preserve consistency and data reliability, as amount of filtered out records are minimal. A combination of Declare Missing Values and No Missing Filter operators were used to achieved this.

To support numerical analysis, non-numerical attributes were filtered (for linear regression) and converted into model-ready formats. Correlation analysis was conducted on numerical predictors to assess relationships with the target variable, while outlier detection was performed using Tukey's method. Outliers outside the acceptable range stated in data dictionary (e.g. Score_ATC only have 0-10 values) were also removed.

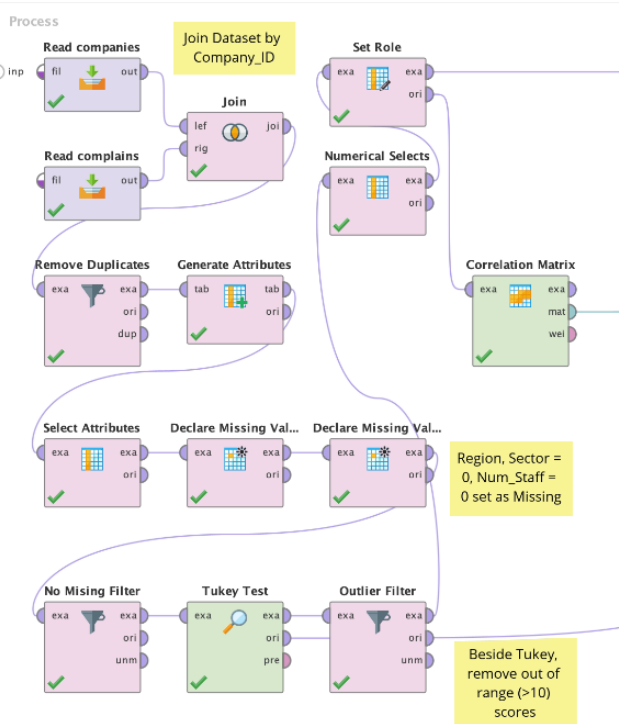


Figure 2.1. RapidMiner Data Preparation Pipeline

overall complaint volume, particularly across finance, hospitality, and technology sectors, suggesting higher density in metropolitan areas. In contrast, Metro-North recorded the lowest complaint volume. Across all regions, finance and hospitality consistently showed higher complaint frequencies than other sectors, suggesting these industries face greater service complexity and customer dissatisfaction risks. This pattern indicates that complaint behaviour is influenced by both operational geography and sectoral characteristics.

Overall, the exploratory analysis identified customer rating and complaint recency as the strongest predictors of regulatory fitness while also highlighting structural complaint concentrating difference between regional and metropolitan, and also across sectors.

Figure 2.1. shows all general data preparation steps conducted in RapidMiner. Additional grouped attributes were later generated specifically for cluster analysis to improve segmentation.

2.2 Distribution, Correlation, and Outlier Analysis

Exploratory analysis was conducted to understand the structural behaviour of the dataset and identify meaningful patterns before predictive modelling.

Correlation analysis (Table 2.2.2) was performed to evaluate relationships between predictor variables and the target variable, Score_ATC. The strongest positive relationship was observed between Avg_Cust_Rating_ATC and Score_ATC ($r = 0.753$), indicating that customer satisfaction is a strong indicator of regulatory fitness performance. Months_Last_Complaint_ATC also showed a strong positive relationship ($r = 0.713$). In comparison, Num_Staff_ATC and Avg_Cust_Trans_Day_ATC showed minimal direct correlation with fitness score, indicating weaker predictive influence.

Complaint distribution analysis (Figure 2.2.1) was conducted to examine the spread of complaint activity across regions and sectors. The results show that Metro-CBD recorded the highest

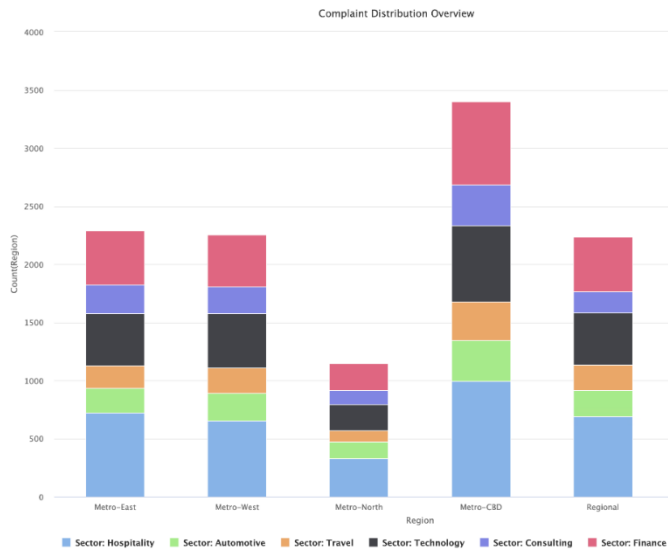


Figure 2.2.1. .Distribution Overview and Boxplot Analysis

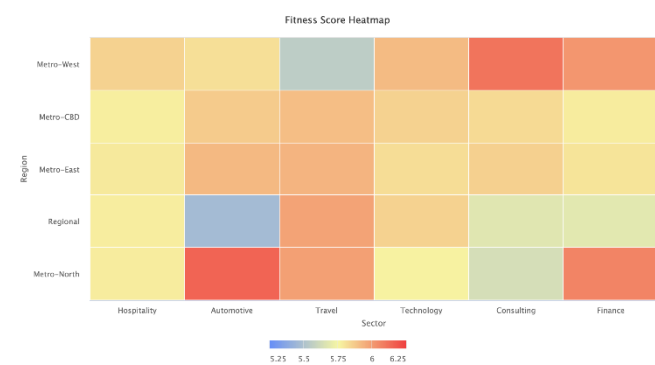


Figure 2.3.2. Region x Sector Fitness Score Heatmap

regional variation, with Metro-North recording the strongest performance while Regional recorded the weakest, suggesting operational inconsistency across locations.

These findings suggest that both geographic and sectoral factors influence regulatory fitness outcomes.

2.4 Cluster Analysis and Pattern Discovery

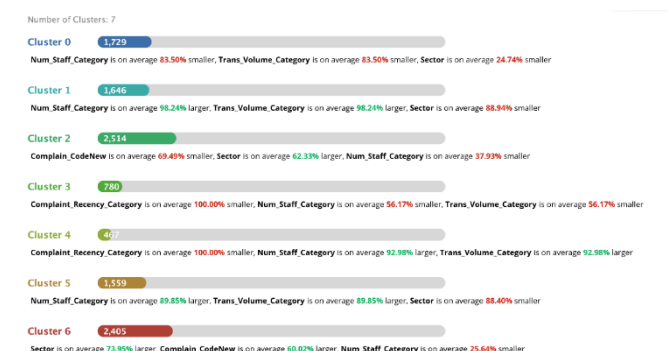


Figure 2.4. Segmentation Profile

were associated with smaller-scale companies and lower transaction intensity.

These results indicate that complaint behaviour differs significantly across operational profiles, supporting more targeted regulatory monitoring based on segment characteristics.

Attributes	Avg_Cust_Rating_ATC	Avg_Cust_Trans_Day_ATC	Months_Last_Complaint_ATC	Num_staff_ATC	Company_ID	Score_ATC ↓
Score_ATC	0.753	0.010	0.713	0.025	0.001	1
Avg_Cust_Rating_ATC	1	0.101	0.727	-0.133	0.013	0.753
Months_Last_Complaint_ATC	0.727	0.024	1	0.022	0.020	0.713
Num_staff_ATC	-0.133	-0.026	0.022	1	-0.006	0.025
Avg_Cust_Trans_Day_ATC	0.101	1	0.024	-0.026	-0.007	0.010
Company_ID	0.013	-0.007	0.020	-0.006	1	0.001

Figure 2.2.2. Correlation Matrix

2.3 Exploratory Pattern Analysis by Region and Sector

To further identify structural business patterns, exploratory analysis was conducted across company region and sector using average fitness score heatmapping (Figure 2.3.2).

The heatmap revealed clear variation in regulatory fitness performance across sector and region combinations, indicating that complaint behaviour and fitness score are not evenly distributed across the market. Metro-West recorded the strongest performance in consulting and finance, showing average fitness scores above 6.0, suggesting stronger complaint management capability in these sectors. In contrast, Metro-North showed the lowest average fitness performance in travel and consulting, indicating comparatively weaker operational fitness in these business environments.

At the sector level, finance and consulting generally maintained stronger fitness score performance across multiple regions, while hospitality and technology showed more moderate and consistent performance. Automotive showed the highest

Cluster analysis was conducted using K-Means clustering to identify segments among complaints. To improve interpretability, three grouped attributes were created specifically for clustering: Num_Staff_Category, Trans_Volume_Category, and Complaint_Recency_Category.

The optimal number of clusters was determined as 7. The cluster profile results (Figure 2.4.) showed clear operational differences. Cluster 2 contained the largest number of complaints (2,514), followed by Cluster 6 (2,405), indicating that a significant proportion share similar characteristics. Clusters 1 and 5 showed higher staff and transaction volume, representing complaints in larger firms, while Clusters 0 and 3

3. Predictive modelling

3.1 Modelling Strategy and Attribute Selection

The primary predictive objective of this study was to estimate the regulator-assigned company fitness score (Score_ATC) based on operational and complaint-related characteristics. As the target variable is continuous and measured on a numerical scale, regression (estimation) modelling was selected as the most appropriate predictive approach.

The predictor attributes selected for modelling were Region, Sector, Complaint_Code, Avg_Cust_Rating_ATC, Avg_Cust_Trans_Day_ATC, Months_Last_Complaint_ATC, and Num_Staff_ATC. These attributes were retained based on both business relevance and exploratory analysis findings from Section 2. Customer rating and complaint recency demonstrated the strongest correlation with the target variable, while region and sector were retained to capture structural business context.

To improve model reliability and reduce dependency on a single data split, cross-validation was implemented as the model validation strategy. This approach allows the training and testing process to be repeated across multiple data partitions, providing a more stable and generalisable estimate of model performance.

Two regression models were selected for this study: Linear Regression and Random Forest Regression. These models were chosen to provide both interpretability and predictive flexibility. Linear Regression provides a transparent baseline model to estimate direct linear relationships, while Random Forest allows for more complex nonlinear interactions and higher predictive adaptability.

3.2 Linear Regression Model Design

The first predictive model was developed using Linear Regression as the baseline regression model. This model was selected because the target variable, Score_ATC, is continuous and the exploratory analysis in Section 2 identified strong direct relationships between several predictor variables and the target outcome, particularly customer rating and complaint recency. Linear Regression provides an interpretable baseline by quantifying the direct effect of each predictor on the fitness score.

The modelling process began with the prepared analytical dataset produced from the preprocessing stage. The Set Role operator was used to assign Score_ATC as the label attribute, establishing it as the prediction target. The Numerical Selects operator was then applied to retain only relevant numeric predictor attributes and remove non-contributing identifiers, including attributes in the equation below.

The model was implemented using Cross-validation with 10 folds. A fixed local random seed of 12345 was used to ensure consistency.

Within the training subprocess, the Linear Regression operator was configured using T-Test feature selection with an alpha level of 0.05, allowing statistically insignificant predictors to be filtered during model construction. A minimum tolerance threshold of 0.05 was applied as the cutoff for identifying high collinearity between predictors.

Within the testing subprocess, the trained model was applied to unseen validation data using the Apply Model operator to generate predicted fitness scores. Following prediction, a residual attribute was generated using the formula: Residual = Actual Score – Predicted Score. Residual generation was included to support deeper model evaluation in the following section, allowing prediction errors to be examined for systematic bias, underestimation, or overestimation patterns across observations.

The final prediction output was then passed to the Performance operator for formal evaluation and later comparison with the Random Forest model.

The final Linear Regression model produced the following estimated regression equation:

$$\widehat{Score}_{ATC} = 1.412 + 0.045(Avg_Cust_Rating_ATC) - 0.001(Avg_Cust_Trans_Day_ATC) + 0.145(Months_Last_Complaint_ATC) + 0.002(Num_Staff_ATC)$$

This regression equation was retained as the final model because all predictor coefficients remained statistically significant under the configured T-test feature selection threshold ($\alpha = 0.05$), with p-values below the significance cutoff. In addition, tolerance values for all predictors remained above the minimum threshold, indicating that multicollinearity was sufficiently controlled and no variables required removal. This confirms that all selected predictors contributed meaningful explanatory value to the final regression structure.

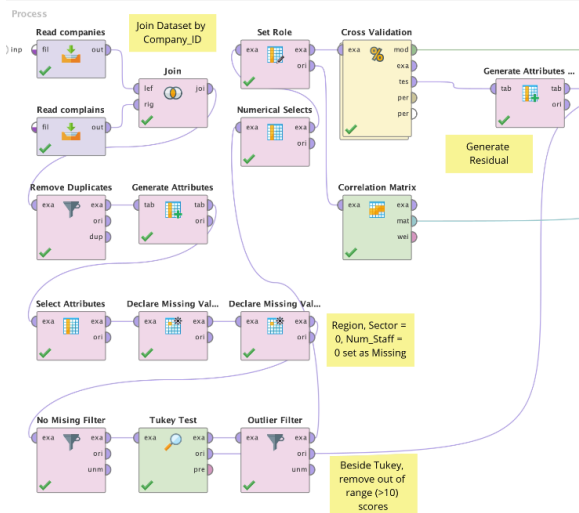


Figure 3.2. Linear Regression RapidMiner Process

3.3 Random Forest Model Design

The second predictive model was developed using Random Forest regression to capture nonlinear relationships and interaction effects between operational variables and the regulator-assigned fitness score (Score_ATC). Unlike the Linear Regression model described in Section 3.2, Random Forest does not rely on linear assumptions and is better suited for modelling more complex predictor behaviour.

The overall modelling pipeline remained the same as the Linear Regression process, including the use of the same prepared dataset, selected predictor attributes, label assignment, residual generation, and 10-fold cross-validation.

Within the training subprocess, the Linear Regression operator was replaced with the Random Forest operator. The model was configured with 100 trees, using least square as the splitting criterion. This criterion was selected because Score_ATC is a continuous numerical target, making squared error minimisation the appropriate split optimisation method for regression modelling. A maximum tree depth of 10 was applied to allow sufficient model complexity while controlling excessive tree growth. The subset ratio was automatically estimated by RapidMiner to improve tree diversity, and a fixed local random seed of 12345 was retained to maintain reproducibility across model runs.

During model tuning, additional parameter variations were tested, including enabling pre-pruning and adjusting tree depth settings. However, these alternative configurations did not improve predictive performance compared to the selected final model. Therefore, the final Random Forest configuration used the simpler non-pre-pruned structure under the selected maximum depth.

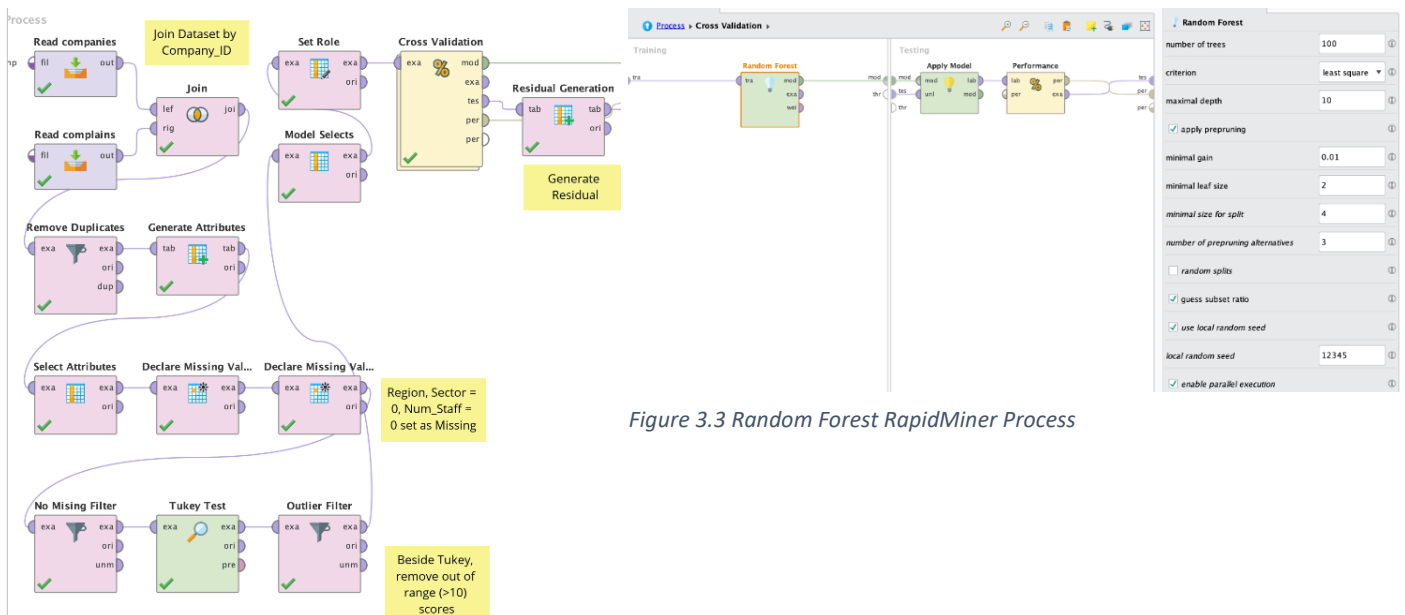


Figure 3.3 Random Forest RapidMiner Process

4. Model evaluation and improvement

4.1 Comparative Model Evaluation

The predictive performance of both developed models was evaluated using the same 10-fold Cross-validation framework to ensure direct comparability under consistent validation conditions. Three regression performance indicators were used for evaluation: Root Mean Squared Error (RMSE), Absolute Error (MAE), and Squared Correlation (R^2).

Given that the target variable (Score_ATC) operates on a bounded scale from 1 to 10, model performance was benchmarked against practical prediction accuracy thresholds. An RMSE below 1.5 was considered acceptable for operational decision-making, while an RMSE closer to 1.0 indicates strong predictive reliability. Similarly, an MAE near 1.0 indicates that predictions deviate by approximately one score unit on average, which is considered operationally useful for regulatory assessment.

Model	RMSE	MAE	R ²
Linear Regression	1.231	1.003	0.634
Random Forest	1.178	0.972	0.665

Table 4.1. Model Performance Comparison

The results show that both models achieved performance within acceptable predictive thresholds. However, Random Forest consistently outperformed Linear Regression across all three performance indicators. Random Forest reduced RMSE by approximately 4.3%, reduced MAE by approximately 3.1%, and improved explanatory fit by increasing R^2 from 0.634 to 0.665.

These results indicate that while both models are operationally usable, Random Forest provides stronger predictive reliability and captures a larger proportion of fitness score variation.

4.2 Residual Analysis and Model Improvement

Residual analysis was performed to further evaluate prediction stability and identify systematic prediction bias. Figure 4.2 presents the residual distribution of the Linear Regression model.

The residual histogram shows an approximately normal distribution centred around zero, indicating that the model does not demonstrate strong systematic overprediction or underprediction. Most prediction errors are concentrated within ± 3 score units, suggesting that the model maintains relatively stable prediction behaviour across the majority of observations.

However, the spread of residuals remains relatively wide, indicating that some observations are not fully captured by the linear structure of the model. This supports the comparative improvement observed in the Random Forest model, which is better able to model nonlinear relationships and interaction effects.

To improve model performance, several tuning variations were tested during Random Forest development, including different tree depth configurations and enabling pre-pruning controls. While these modifications improved model constraint and generalisation control, they did not improve overall predictive error performance compared to the final selected model. As a result, the final Random Forest model retained the stronger-performing parameter structure.

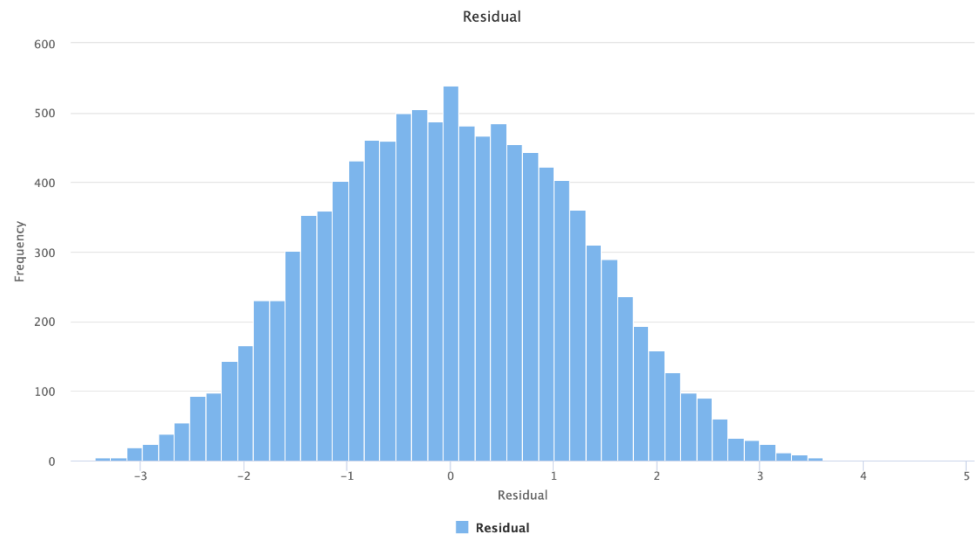


Figure 4.2. Linear Regression Residual Distribution

4.3 Association Analysis: Frequently Co-occurring Complaints

To identify systemic complaint behaviour across companies, Association Rule Learning was applied to company-level complaint baskets, where complaint codes were transformed into binary occurrence records. A minimum support of 0.1 in FP-Growth, and a lift threshold of 7 was used to retain only high-strength complaint associations.

No.	Premises	Conclusion	...	Confidence	LaPlace	Gain	p-s	Lift	Convic...
3	CC-721	CC-774	...	0.509	0.967	-0.106	0.031	7.184	1.894
4	CC-721	CC-921	...	0.509	0.967	-0.106	0.031	7.253	1.895
5	CC-721	CC-774, CC-921	...	0.509	0.967	-0.106	0.034	13.869	1.964
6	CC-774	CC-721	...	0.512	0.968	-0.105	0.031	7.184	1.904
7	CC-774	CC-721, CC-921	...	0.512	0.968	-0.105	0.034	14.102	1.976
8	CC-921	CC-721	...	0.517	0.968	-0.104	0.031	7.253	1.924
9	CC-921	CC-721, CC-774	...	0.517	0.968	-0.104	0.034	14.238	1.996
10	CC-774	CC-921	...	0.518	0.968	-0.105	0.032	7.375	1.929
11	CC-921	CC-774	...	0.523	0.969	-0.104	0.032	7.375	1.948
12	CC-774, CC-921	CC-721	...	0.989	1.000	-0.037	0.034	13.869	84.510
13	CC-834, CC-972	CC-361	...	0.993	1.000	-0.039	0.034	8.870	126.991
14	CC-834, CC-253	CC-361	...	0.996	1.000	-0.037	0.033	8.900	245.102
15	CC-721, CC-774	CC-921	...	1	1	-0.036	0.034	14.238	∞
16	CC-721, CC-921	CC-774	...	1	1	-0.036	0.034	14.102	∞

Table 4.3. Top Co-occurring Complaint Rules by Lift

The analysis identified three high-strength recurring complaint patterns across companies. The strongest complaint cluster involved CC-721, CC-774, and CC-921, where the co-occurrence of CC-721 and CC-774 consistently resulted in CC-921 (confidence = 1, lift = 14.238). This indicates that these complaint types form a tightly connected complaint ecosystem and are highly likely to stem from the same underlying operational issue.

A second recurring pattern was identified between CC-834, CC-253, and CC-361, where the presence of CC-834 and CC-253 almost always resulted in CC-361 (confidence = 0.996, lift = 8.9). This suggests a highly stable complaint pathway, indicating that these complaint issues are strongly interdependent rather than isolated incidents.

A third pattern linked CC-834, CC-972, and CC-361, where CC-834 and CC-972 also showed a near-complete association with CC-361 (confidence = 0.993, lift = 8.87). The repeated appearance of CC-834 and CC-361 across multiple strong rules suggests that these complaint types may act as core indicators of a broader recurring operational problem.

These findings show that complaint behaviour is structured around recurring complaint ecosystems rather than isolated complaint events. For the regulator, this provides a stronger foundation for policy design by designing targeted intervention at the complaint-cluster level. Companies exhibiting one or more complaints within these clusters can be prioritised for earlier compliance review, improving preventative action and reducing the likelihood of repeated complaint escalation.

4.4 Decision-Making Implications

The evaluation results suggest that both predictive models can support regulatory decision-making by estimating company fitness scores from complaint and operational characteristics. The Linear Regression model provides stronger interpretability, allowing the regulator to understand direct predictor influence, particularly customer rating and complaint recency.

However, the Random Forest model provides stronger predictive accuracy and better explanatory coverage, making it the preferred model for operational deployment. Its lower prediction error means that company fitness estimates can be generated more reliably, improving the regulator's ability to prioritise investigations, allocate compliance resources, and identify companies with elevated complaint risk.

In practice, the Random Forest model can be integrated into complaint assessment workflows as an early warning analytical tool, while the Linear Regression model can continue to support interpretative policy analysis and variable-level business understanding.